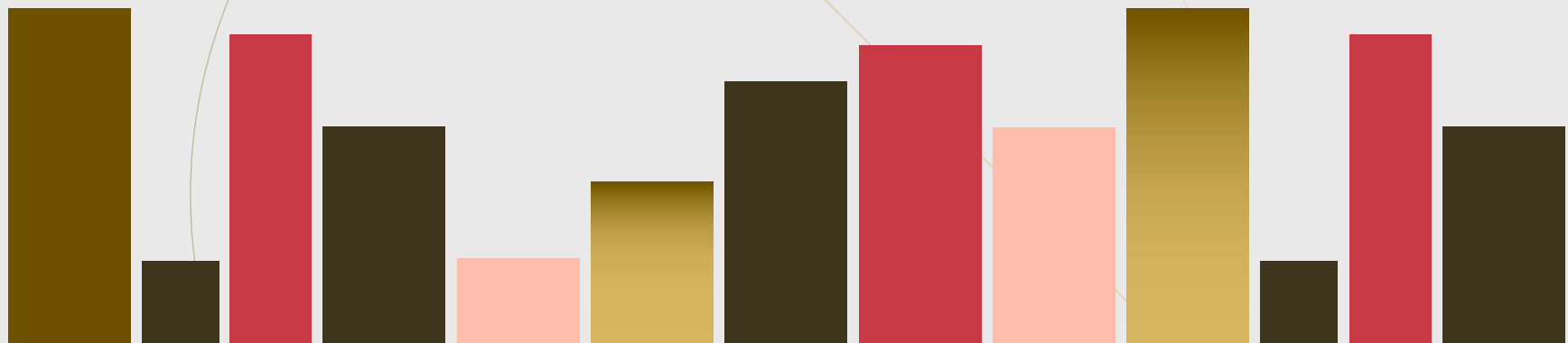


ZEN IUT1-8th



7. Define $b_k = 1 + a + \dots + a^k = \sum_{i=0}^k a^i$

The statement is $\lim_{k \rightarrow \infty} b_k = \frac{1}{1-a}$

$b_k = \frac{1 - a^{k+1}}{1 - a}$ (This formula is true for any field by mathematical induction)

$$\left| \frac{1}{1-a} - b_k \right|_p = \left| \frac{a^{k+1}}{1-a} \right|_p = \frac{|a|_p^{k+1}}{|1-a|_p} \rightarrow 0 \quad \text{when } k \rightarrow \infty$$

$\wedge (|a| < 1)$ ■

8.

$$m_1 = m_2 = 2$$

$$\gamma_4(m_1, m_2) \neq \gamma_4(m_1) + \gamma_4(m_2)$$

$$\begin{array}{c} \parallel \\ \gamma_4(4) \end{array}$$

 $\parallel$

4

$$\begin{array}{c} \parallel \\ \gamma_4(2) + \gamma_4(2) \end{array}$$

 $\parallel$

1 + 1

 $\parallel$

2



9. Prop. For odd prime p ,

$X^2 + 1$ has a root in $\mathbb{Q}_p$

$(\Leftrightarrow)$

-1 is a quadratic residue in mod p .

Proof.

$$(\Rightarrow) \quad X^2 = -1 \quad (\in \mathbb{Q}_p)$$

Take mod p both sides,

$$\text{We'll get } X^2 \equiv -1 \pmod{p}$$

$$4p+3 \rightarrow -1$$

$$4pt \rightarrow 1$$

($\Leftarrow$) Define $f(x) = x^2 + 1$, $f'(x) = 2x$

Take $a^2 \equiv -1 \pmod{p}$, $f(a) \equiv 0$, $f'(a) \not\equiv 0 \pmod{p}$

($p \geq 3$, $2 \not\equiv 0$)

Applying Hensel's lemma, we can get a root of $f(x)$
in $\mathbb{Q}_p$. $\blacksquare$

For $p=2$, $-1 = 1 + 2 + 2^2 + \dots$

If $a^2 = -1$ for some a ,

take mod 4 both sides.

$$a^2 \equiv 1 + 2$$

but 3 in mod 4 is quadratic non-residue.

Therefore, there's no root of $X^2 + 1$ in $\mathbb{Q}_2$.

For odd primes, $(4p+1)$ -formed primes only have a root.

p	2	3	5	11	11	13	17	19
root	X	X	0	X	X	0	0	X



10. Assume $\varphi: \mathbb{Z}_p \rightarrow \mathbb{Z}_p$ is isomorphism.

$$\varphi(1) = 1$$

$$\varphi\left(q \cdot \frac{1}{q}\right)$$

$$\varphi(q) \cdot \varphi\left(\frac{1}{q}\right)$$

$$\frac{q}{p} \cdot \varphi\left(\frac{1}{q}\right)$$

$$\frac{1}{q} \in \mathbb{Z}_p \quad \hookrightarrow \quad \left(\frac{1}{q}\right)_p = 1$$

Prop. For $n \in \mathbb{N}$, $\varphi(n) = n$

First, $\varphi(1) = 1$

$$\varphi(2) = \varphi(1+1) = \varphi(1) + \varphi(1) = 2$$

And continuing, $\varphi(n) = n$.

$$q \cdot \varphi\left(\frac{1}{q}\right) = 1$$

Take $|\cdot|_q$ both sides,

$$|q|_q \left| \varphi\left(\frac{1}{q}\right) \right|_q = |1|_q$$

$$\left| \varphi\left(\frac{1}{q}\right) \right|_q = q$$

but φ 's codomain is $\mathbb{Z}_q^\times$

$$\{\alpha \mid \alpha \leq 1\}$$

"

11.

$$x \in 1 + p \mathbb{Z}_p \Leftrightarrow \forall m > 0, p \nmid m \Rightarrow \exists a, a^m = x.$$

Pf

$$(\Rightarrow) \text{ Let } x = 1 + a_1 p + a_2 p^2 + \dots \quad (0 \leq a_i < p)$$

$$\text{Consider } y = b_0 + b_1 p + b_2 p^2 + \dots \text{ to solve } x = y^m.$$

$$\text{By mod } p, \quad 1 \equiv b_0$$

$$\text{mod } p^2, \quad a_1 \equiv \underline{mb_1} \quad \text{with } mC_1$$

$$\text{mod } p^3, \quad a_2 \equiv mb_2 + mC_2 b_1$$

⋮

Generally, let $f_n(x) = b_0 + b_1x + b_2x^2 + \dots + b_nx^n$

and denote $[x^k]g$ as the coefficient of x^k of polynomial g .

$$\text{mod } p, \quad a_k \equiv m b_k + [x^k]f_{k-1}^m \quad (k \geq 1)$$

Here, $p \nmid m$ so b_k can be solvable.

By this algorithm, y is decidable.

$$(\Leftrightarrow) \quad \text{Let } |a|_p = p^k, \quad |x|_p = p^s \quad (k, s \in \mathbb{Z})$$

$$a^m = x \quad \Rightarrow \quad |a|_p^m = |x|_p \quad \Leftrightarrow \quad p^{km} = p^s$$

$$\Leftrightarrow \underline{km = s} \quad (*)$$

$p \nmid np-1$ so it can be m , especially any larger integers.

If $|x|_p \neq 1$, then $s \neq 0$ and also $k \neq 0$.

Take $m > |s|$, and this contradicts with $(*)$

Thus, $|x|_p = 1$.

Let x be $a_0 + a_1 p' + a_2 p'^2 + \dots$ ($0 \leq a_i < p$, $a_i \in \mathbb{Z}$)

Take m as $(p-1)$ again, and take mod p , for some b ,

by
FLT $\hookrightarrow$

$$b^{p-1} \equiv a_0 \pmod{p}$$

Finally we get x is $1 + a_1 p' + a_2 p'^2 + \dots$ formed.

This generates $1 + p\mathbb{Z}_p$. $\blacksquare$